

Wrench-Market Games: Distributed Coalition Formation and Certified Cooperative Transport for Heterogeneous Mobile Robots

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Abstract

We develop a layered game-theoretic theory for the recruitment of heterogeneous mobile robots into coalitions that cooperatively transport heterogeneous loads requiring both force and torque. At the recruitment layer, we prove that the deficit-driven population game loses its exact potential structure under capacity heterogeneity, and that the structure is restored—exactly, and only—by measuring the population state in capacity units, or equivalently by pricing each robot’s *marginal effective capacity* in the payoff; Smith-type revision dynamics then ascend an explicit Lyapunov potential, and an integer quorum-clearing layer with a potential guard converges in finitely many events with an explicit switching budget and a rounding-loss bound. At the physical layer, the coalition’s reachable wrench set is a *constant body-frame zonotope*; feasibility is a linear program and a capacity margin is computable via support functions; we prove that cardinality and scalar capacity criteria are structurally insufficient, that the wrench value exhibits strict complementarity invalidating diminishing-marginal-gain auction guarantees (with an unbounded approximation-ratio lower bound), and a *triple identity* equating the market-clearing wrench price, the contact-constraint Lagrange multiplier, and the supporting facet normal of the zonotope. At the control layer, we derive a coupled control law from a single strategic–mechanical potential and prove—without timescale separation—that the interconnection of evolutionary strategy revision and Euler–Lagrange motion converges to a new equilibrium concept, the *Nash–Lagrange equilibrium*; receding-horizon transport with a shared tracking cost is shown to be an *exact dynamic potential game*, so distributed Nash-seeking recovers the team optimum, and high-order control-barrier constraints are half-spaces in wrench space, preserving the geometric certificate machinery for safety. An endogenous communication layer is analyzed: information is priced in physical units, connectivity provision is a public good subject to free riding, and pairwise-comparison dynamics select the comparative-advantage equilibrium in which the lightest robots specialize as mobile relays. The theory’s falsifiable predictions are validated in audited Monte Carlo campaigns (more than 50,000 evaluations, zero audit failures), including a payoff correction prescribed by the theory that improves the corresponding distributed rule exactly in the predicted direction.

Keywords: population games, potential games, multi-robot task allocation, coalition formation, cooperative manipulation, distributed control, control barrier functions.

1 Introduction

Mature intralogistics automation moves standardized load units with individual autonomous mobile robots (AMRs) coordinated by fleet managers. This paradigm breaks when loads are heavy, long, fragile, or unbalanced: such loads demand *coalitions*—several robots applying forces at several contact points—and the correct coalition is determined not by head-count or aggregate payload, but by the geometry of achievable forces and torques. Three bodies of literature meet at this problem and each leaves a structural gap. Multi-robot task allocation and coalition formation [1, 2, 3] treat *who works with whom* but carry no dynamics, no contact geometry, and no control-theoretic certificates. Cooperative manipulation [4, 5, 6] controls a load with a *given*

team, typically homogeneous, and does not decide recruitment. Generalized Nash equilibrium (GNE) seeking in population games [7, 8, 9, 10] provides distributed protocols with convergence guarantees, but on a continuum of agent mass, explicitly abstracting integrality, and with no physical feasibility notion beyond scalar constraints.

This paper closes the loop between these layers with two unifying objects. The first is the coalition’s *reachable wrench set* $\mathcal{W}(C)$ —a zonotope, constant in the load’s body frame—which acts successively as the recruitment certificate, the full-actuation condition, the input-constraint set of the transport controller, and the support of the safety filter. The second is the *wrench price* $\mu \in \mathbb{R}^3$, which is simultaneously the dual variable of a minimum-effort force market, the Lagrange multiplier of the rigid contact constraint, the supporting facet normal of $\mathcal{W}(C)$ at saturation, and (in the predictive layer) the costate of the transport optimal control problem.

Contributions. (i) A complete characterization of when the heterogeneous recruitment game is a potential game (Lemma 1, Theorems 1–2): exact potential structure holds *iff* the state is measured in capacity units, or equivalently *iff* payoffs price marginal effective capacity—turning a design slogan (“loads buy capacity, not robots”) into a necessary and sufficient condition. (ii) Convergence of the full hybrid architecture: Smith-type dynamics ascend the potential (Theorem 3); integer clearing with a potential guard terminates in finitely many events with an explicit switching budget and a rounding bound vanishing with polarization (Theorem 4, Corollary 1, Proposition 1). (iii) Impossibility results for the physical layer: cardinality and scalar capacity are insufficient (Proposition 3); the wrench value is complementary, so diminishing-marginal-gain (DMG) auction guarantees do not apply, with an *unbounded* approximation-ratio lower bound (Proposition 4, Theorem 5); and the continuous margin dissolves the impossibility by restoring marginal signal (Remark 3). (iv) The triple price identity (Theorem 6) and the internal-force decomposition with zero market price (Proposition 6). (v) A control law derived from a single strategic–mechanical potential whose closed loop—strategy revision coupled to Euler–Lagrange motion—is certified by one energy function and converges to Nash–Lagrange equilibria without timescale separation (Theorem 7); trajectory-tube recruitment and margin-certified tracking (Section 8.3); an exact dynamic potential game result making distributed predictive transport team-optimal (Theorem 9); and safety constraints as wrench-space half-spaces (Proposition 8). (vi) An endogenous-topology analysis: connectivity priced in physical units, a free-riding impossibility and its repair by common-value potential design, and an equilibrium-selection result: pairwise comparison dynamics select the comparative-advantage outcome in which the lightest robots become mobile communication hubs (Section 9). (vii) Audited numerical validation of the theory’s falsifiable predictions (Section 10).

2 Problem formulation and standing assumptions

Let $\mathcal{R} = \{1, \dots, N\}$ be the robots and $\mathcal{L} = \{1, \dots, M\}$ the loads; $\mathcal{L}_0 = \mathcal{L} \cup \{0\}$ includes the idle strategy. Robot i holds a mixed/portfolio state $x_i \in \Delta_i = \{x \in \mathbb{R}_{\geq 0}^{M+1} : \sum_k x_{ik} = 1\}$; the joint state is $x \in \mathcal{X} = \prod_i \Delta_i$. Robot i has nominal capacity $c_i > 0$; the capacity allocated to load k is $S_k(x) = \sum_i c_i x_{ik}$, and $C_{\text{tot}} = \sum_i c_i$.

Assumption 1. *The recruitment payoff is*

$$p_{ik}(x) = V_k \sigma(D_k - S_k(x)) - g_{ik}, \quad p_{i0} \equiv 0, \quad (1)$$

with load value $V_k > 0$, demand $D_k > 0$, $\sigma : \mathbb{R} \rightarrow (0, 1)$ of class C^1 , strictly increasing with $0 < \sigma' \leq L_\sigma$, and separable costs $g_{ik} \geq 0$ (distance, energy, battery) independent of x ; $\bar{g}_i = \max_k g_{ik}$, $V_{\max} = \max_k V_k$. All strategic coupling enters through the aggregates S_k (congestion or over-assignment terms, if present, are functions of S_k).

Assumption 2. *Each robot obeys Euler–Lagrange dynamics $M_i(q_i)\ddot{q}_i + C_i(q_i, \dot{q}_i)\dot{q}_i + g_i(q_i) = \tau_i + J_i(q_i)^\top f_i$ with $M_i \succ 0$ and $\dot{M}_i - 2C_i$ skew-symmetric; the workspace $\mathcal{Q}$ is compact. At the*

contact layer, each slot force set $U_h \subset \mathbb{R}^2$ is convex, compact, with $0 \in U_h$ (disks for bilateral grasps; truncated friction cones for unilateral pushing). Robots can exert planar forces in any direction at their contact; nonholonomic force cones are discussed in Section 11.

Assumption 3. Sections 3–8.3 assume the aggregates S_k (and, where relevant, the load state) are known to the deciding robots; local estimation over time-varying graphs is treated in Section 9 and quantified empirically in the validation campaigns.

3 Potential structure of heterogeneous recruitment

Lemma 1 (Obstruction under heterogeneity). *Under Assumption 1, if $c_i \neq c_j$ for some $i \neq j$, then no $\Phi \in C^2(\mathcal{X})$ satisfies $\partial\Phi/\partial x_{ik} = p_{ik}$ for all i, k : the game is not an exact (unweighted) potential game [11, 9].*

Proof. Existence of Φ requires symmetry of second derivatives on the convex domain $\mathcal{X}$: $\partial p_{ik}/\partial x_{jk} = \partial p_{jk}/\partial x_{ik}$. From (1), $\partial p_{ik}/\partial x_{jk} = -V_k \sigma'(D_k - S_k) c_j$ and $\partial p_{jk}/\partial x_{ik} = -V_k \sigma'(D_k - S_k) c_i$. Since $V_k > 0$ and $\sigma' > 0$ everywhere, symmetry forces $c_i = c_j$. $\square$

Theorem 1 (Capacity-weighted exact potential). *Define*

$$\Phi(x) = \sum_{k \in \mathcal{L}} V_k \Psi_k(S_k(x)) - \sum_{i \in \mathcal{R}} \sum_{k \in \mathcal{L}} c_i g_{ik} x_{ik}, \quad \Psi_k(S) = \int_0^S \sigma(D_k - s) ds. \quad (2)$$

Then (i) $\partial\Phi/\partial x_{ik} = c_i p_{ik}(x)$ for all $i \in \mathcal{R}, k \in \mathcal{L}_0$: the game is an exact weighted potential game with weights $w_i = c_i$; (ii) equivalently, in capacity coordinates $z_{ik} = c_i x_{ik}$ (each robot a population of mass c_i [7, Rem. 2.1]) the game is an exact unweighted potential game with $\partial\Phi/\partial z_{ik} = p_{ik}$; (iii) an exact unweighted potential in x -coordinates exists iff $c_1 = \dots = c_N$.

Proof. (i) For $k \in \mathcal{L}$: $\partial\Phi/\partial x_{ik} = V_k \sigma(D_k - S_k) c_i - c_i g_{ik} = c_i p_{ik}$; for $k = 0$ both sides vanish. (ii) With $S_k = \sum_i z_{ik}$, $\partial\Phi/\partial z_{ik} = V_k \sigma(D_k - S_k) - g_{ik} = p_{ik}$. (iii) Sufficiency: divide (i) by the common capacity; necessity is Lemma 1. $\square$

When effective capacity depends on the robot–load pair, $e_{ik} = c_i \beta(b_i) \exp(-\|q_i - p_k\|/\ell_d)$ (battery- and horizon-discounted deliverable capacity), even the weighted structure fails—unless the payoff prices *marginal* capacity:

Theorem 2 (Marginal-capacity pricing). *Let $e_{ik} > 0$ be fixed at decision time and $S_k(x) = \sum_i e_{ik} x_{ik}$. Then: (i) with payoffs*

$$p_{ik}(x) = e_{ik} V_k \sigma(D_k - S_k(x)) - g_{ik}, \quad (3)$$

the game is an exact potential game in the original coordinates, with $\Phi(x) = \sum_k V_k \Psi_k(S_k) - \sum_{i,k} g_{ik} x_{ik}$ and $\partial\Phi/\partial x_{ik} = p_{ik}$; (ii) with the plain payoff $p_{ik} = V_k \sigma(D_k - S_k) - g_{ik}$, a weighted potential with weights $\{w_i\}$ exists iff e_{ik} factorizes as $e_{ik} = w_i \varphi_k$; in particular, non-separable distance discounts admit no (weighted) potential. Theorem 1 is the case $\varphi_k \equiv 1$, $w_i = c_i$.

Proof. (i) Direct differentiation: $\partial\Phi/\partial x_{ik} = V_k \sigma(D_k - S_k) e_{ik} - g_{ik} = p_{ik}$. (ii) Symmetry of $\partial(w_i p_{ik})/\partial x_{jk}$ requires $w_i e_{jk} = w_j e_{ik}$ for all i, j, k , i.e., $e_{ik}/w_i = e_{jk}/w_j =: \varphi_k$, hence the factorization; for sufficiency, $\Phi(x) = \sum_k (V_k/\varphi_k) \Psi_k(S_k) - \sum_{i,k} w_i g_{ik} x_{ik}$ verifies $\partial\Phi/\partial x_{ik} = w_i p_{ik}$. $\square$

Remark 1. *Theorem 2 turns the design principle “the load buys marginal capacity, not presence” into the unique potential-preserving payoff within this family. It generates a falsifiable prediction—a plain-payoff rule should disperse capacity and lose coalition completion—confirmed experimentally in Section 10.*

4 Evolutionary dynamics and convergence

Each robot revises by Smith dynamics [12]

$$\dot{x}_{ik} = \sum_{\ell \in \mathcal{L}_0} x_{i\ell} [p_{ik} - p_{i\ell}]_+ - x_{ik} \sum_{\ell \in \mathcal{L}_0} [p_{i\ell} - p_{ik}]_+. \quad (4)$$

Lemma 2 (Positive correlation). *For every i and every $x \in \mathcal{X}$,*

$$\sum_k p_{ik} \dot{x}_{ik} = \sum_{k,\ell} x_{i\ell} ([p_{ik} - p_{i\ell}]_+)^2 \geq 0, \quad (5)$$

with equality iff every used strategy of i is payoff-maximal (best response); rest points of (4) coincide with the latter.

Proof. Substituting (4), relabeling $k \leftrightarrow \ell$ in the outflow sum, and using $a[a]_+ = ([a]_+)^2$: $\sum_k p_{ik} \dot{x}_{ik} = \sum_{k,\ell} x_{i\ell} (p_{ik} - p_{i\ell}) [p_{ik} - p_{i\ell}]_+ = \sum_{k,\ell} x_{i\ell} ([p_{ik} - p_{i\ell}]_+)^2$. Vanishing of the sum is equivalent to $x_{i\ell} > 0 \Rightarrow p_{ik} \leq p_{i\ell} \forall k$; in that case both terms of (4) vanish; conversely $\dot{x}_i = 0$ annihilates the left side. $\square$

Theorem 3 (Global convergence to Nash). *Under Assumptions 1 and 3 (with either the weighted potential of Theorem 1 or the marginal payoff of Theorem 2), solutions of (4) remain in $\mathcal{X}$, Φ is nondecreasing with $\dot{\Phi} = \sum_i w_i \sum_{k,\ell} x_{i\ell} ([p_{ik} - p_{i\ell}]_+)^2 \geq 0$ ($w_i = c_i$ or 1 respectively), and $\text{dist}(x(t), \text{NE}) \rightarrow 0$.*

Proof. The field (4) is Lipschitz on the compact $\mathcal{X}$ (payoffs C^1 , $[\cdot]_+$ Lipschitz); the simplex is invariant since outflow vanishes on faces and $\sum_k \dot{x}_{ik} = 0$. By Theorems 1/2 and the chain rule, $\dot{\Phi} = \sum_i w_i (p_i \cdot \dot{x}_i)$, nonnegative by Lemma 2. LaSalle's invariance principle on $\mathcal{X}$ yields convergence to the largest invariant set in $\{\dot{\Phi} = 0\}$, which by Lemma 2 is exactly the set of rest points = Nash equilibria. $\square$

Remark 2. *No rates are claimed, convergence is to the Nash set, and the result holds for any revision protocol with positive correlation (BNN, replicator, projected gradient): the guarantee belongs to the potential, not to Smith specifically.*

5 Hybrid integer clearing

Transport requires integer coalitions. Let $A : \mathcal{R} \rightarrow \mathcal{L}_0$ with coalitions $C_k(A)$; A is *admissible* if every nonempty coalition satisfies its quorum predicate Q_k (cardinality $|C_k| \geq n_k$, or wrench feasibility of Section 6); the admissible set $\mathcal{A}$ is finite. Let Φ_d be the restriction of (2) to vertices. At event times, a proposal map generates a candidate $A' \in \mathcal{M}(A) \subseteq \mathcal{A}$ (containing at least all admissible unilateral reassignments and dissolutions); the *potential guard* accepts iff $\Phi_d(A') \geq \Phi_d(A) + \delta$ ($\delta \geq 0$; $\delta = 0$ means strict increase).

Theorem 4 (Finite convergence to quorum-stable assignments). *The number of accepted changes is at most $|\mathcal{A}| - 1 \leq (M+1)^N - 1$; after the last event the assignment A^* is quorum-stable: no move in $\mathcal{M}(A^*)$ increases Φ_d , and A^* is a Nash equilibrium of the discrete game restricted to $\mathcal{A}, \mathcal{M}$.*

Proof. Accepted changes strictly increase Φ_d over a finite set, so no assignment repeats (finite-improvement property [11]); termination at a local maximum of Φ_d under $\mathcal{M}$ follows from the guard's definition. A profitable unilateral deviation of robot i would raise the weighted potential by $c_i \times (\text{gain}) > 0$ (the variation of Φ_d between vertices differing only in i integrates $c_i p_i$ along the segment) and lies in $\mathcal{M}(A^*)$, contradiction. $\square$

Corollary 1 (Switching budget). *With hysteresis $\delta > 0$, the number of accepted switches satisfies $N_{\text{sw}} \leq (\Phi_{\text{max}} - \Phi_{\text{min}})/\delta \leq (V_{\text{max}}C_{\text{tot}} + \sum_i c_i \bar{g}_i)/\delta$, since $0 \leq \Psi_k(S) \leq S$ and $\sum_k S_k \leq C_{\text{tot}}$.*

Proposition 1 (Rounding loss). *Let $\hat{x}$ be the argmax rounding of x and $\eta_i = 1 - \max_k x_{ik}$. Then $|\Phi(x) - \Phi(\hat{x})| \leq 2 \sum_i c_i (V_{\text{max}} + \bar{g}_i) \eta_i$; in particular the discretization loss vanishes as the continuous phase polarizes ($\eta_i \rightarrow 0$), which is precisely what (4) with memory produces.*

Proof. $\sum_k |x_{ik} - \hat{x}_{ik}| = 2\eta_i$, hence $\sum_k |S_k(x) - S_k(\hat{x})| \leq 2 \sum_i c_i \eta_i$; the deficit block of Φ is 1-Lipschitz in each S_k scaled by V_k ($\|\sigma\|_\infty \leq 1$), and the separable block is bounded by $2 \sum_i c_i \bar{g}_i \eta_i$. $\square$

6 Wrench geometry: certificates and impossibility results

Definition 1 (Contact map, reachable set, margin). *Slots $r_h \in \mathbb{R}^2$ in the load frame define contact maps $A_h u = (u_x, u_y, r_{hx}u_y - r_{hy}u_x)^\top \in \mathbb{R}^3$. The coalition's reachable wrench set is the Minkowski sum $\mathcal{W}(C) = \bigoplus_{h \in h(C)} A_h U_h$ (a zonotope for polytopic U_h). Fixing a wrench norm $\|(f, \tau)\| = \|f\|_2 + |\tau|/L_c$ (L_c a characteristic load length), the capacity margin is $\rho(C, w) = \max\{r : w + r\mathbb{B} \subseteq \mathcal{W}(C)\}$ (signed by penetration for $w \notin \mathcal{W}$).*

Proposition 2 (LP test and support-function margin). *With $h_{\mathcal{W}}(\eta) = \sum_h h_{U_h}(A_h^\top \eta)$ ($h_U(v) = \sup_{u \in U} v^\top u$; $h_U(v) = \bar{f}\|v\|_2$ for disks): (i) $w \in \mathcal{W}(C) \iff \eta^\top w \leq h_{\mathcal{W}}(\eta) \forall \eta$, an LP for polytopic U_h ; (ii) $\rho(C, w) = \min_{\|\eta\|_* = 1} (h_{\mathcal{W}}(\eta) - \eta^\top w)$, attained on the finite set of facet normals for zonotopes [16, 17].*

Proof. (i) is support-function separation together with $h_{\bigoplus_h A_h U_h} = \sum_h h_{U_h} \circ A_h^\top$. (ii) $w + r\mathbb{B} \subseteq \mathcal{W} \iff \eta^\top w + r\|\eta\|_* \leq h_{\mathcal{W}}(\eta) \forall \eta$; taking the supremum over r gives the stated minimum, piecewise linear in η , hence attained at facet normals. $\square$

Proposition 3 (Cardinality and scalar capacity are insufficient). *For every n and every total capacity, there exist load geometries, demands and coalitions with $|C| \geq n$ and $\sum_{i \in C} c_i \geq D_k$ but $w^{\text{dem}} \notin \mathcal{W}(C)$: (a) directional obstruction: if all accessible slots push through cones about a common inward normal n_0 ($u^\top n_0 \geq \kappa \|u\|$, $\kappa > 0$), every achievable wrench has $f^\top n_0 \geq 0$, so any demand with $f^{\text{dem}\top} n_0 < 0$ is infeasible for all coalitions; (b) rotational saturation: if $\|r_h\| \leq a$ and $\|u_h\| \leq c_i$, then $|\tau| \leq a \sum_{i \in C} c_i$, so $|\tau^{\text{dem}}| > aC_{\text{tot}}$ is infeasible although the scalar test passes.*

Proof. (a) $f^\top n_0 = \sum_h u_h^\top n_0 \geq \kappa \sum_h \|u_h\| \geq 0$. (b) $|\tau| \leq \sum_h \|r_h\| \|u_h\|$ by Cauchy–Schwarz termwise. $\square$

Proposition 4 (Strict complementarity of the wrench value). *Let $v(C) = \mathbf{1}\{w^{\text{dem}} \in \mathcal{W}(C)\}$. Then v is monotone (since $0 \in U_h$) but not submodular: for a bar with slots at $r_{1,2} = (\pm a, 0)$, disks of radius $\bar{f}$, and pure-torque demand $w^{\text{dem}} = (0, 0, \tau_d)$, $0 < \tau_d \leq 2a\bar{f}$,*

$$v(\emptyset) = v(\{1\}) = v(\{2\}) = 0, \quad v(\{1, 2\}) = 1,$$

violating diminishing marginal returns at $A = \emptyset \subset B = \{2\}$. Indeed a single contact can never realize a nonzero pure torque: the demand forces net force $u = 0$, hence $\tau = 0$.

Theorem 5 (Unbounded inapproximability of DMG auctions). *For every n there is an instance (n pure-torque bars, $2n$ robots) on which any mechanism whose bids equal the individual marginal of the feasibility indicator—the diminishing-marginal-gain scoring class that includes canonical CBBA [14, 15]—serves 0 loads while the optimum serves n : the approximation ratio of the class is unbounded.*

Proof. By Proposition 4, every singleton marginal on an empty pure-torque load is zero; all bids are zero, no coalition is initiated, and the empty assignment is a fixed point. The optimum pairs robots at opposite ends of each bar. $\square$

Remark 3 (The margin dissolves the impossibility). *The impossibility is a property of the 0/1 indicator, not of the problem: the continuous residual strictly decreases even for the first robot on a pure-torque load (a force at $(a, 0)$ can trade force error against torque error), so margin-based signals restore marginal gradient. This predicts that (i) auction engines equipped with a continuous wrench signal escape Theorem 5, and (ii) marginal-margin $(\Delta\rho)$ signals remain myopic relative to deficit-margin signals under complementarity—both confirmed in Section 10.*

Proposition 5 (Scalar model as exact degenerate case). *If all slots are collocated at the center of mass ($r_h = 0$) with $U_h = \{\|u\|_2 \leq c_i\}$, then $\mathcal{W}(C) = \{(f, 0) : \|f\|_2 \leq \sum_{i \in C} c_i\}$ and, for pure-force demands, wrench feasibility is exactly the scalar test $\sum_{i \in C} c_i \geq \|f^{\text{dem}}\|_2$. Propositions 3–4 characterize precisely what this projection loses: directionality, leverage, and complementarity.*

7 Force allocation and the triple price identity

Fix a coalition with stacked grasp map $G \in \mathbb{R}^{3 \times 2H}$ of full row rank and effort metric $R = \text{blkdiag}(R_h) \succ 0$.

Theorem 6 (Triple identity of the wrench price). *Consider (a) the coordinator problem $\min_\lambda \frac{1}{2} \lambda^\top R \lambda$ s.t. $G\lambda = w^{\text{dem}}$, and (b) the market in which each contact best-responds to a price $\mu \in \mathbb{R}^3$: $\lambda_h(\mu) = \arg \max_{\lambda_h} \{\mu^\top A_h \lambda_h - \frac{1}{2} \lambda_h^\top R_h \lambda_h\}$, with μ clearing $G\lambda(\mu) = w^{\text{dem}}$. Then, at interior optima:*

- (i) *the competitive equilibrium exists, is unique, and equals the coordinator optimum: $\lambda^* = R^{-1} G^\top \mu^*$, $\mu^* = (GR^{-1} G^\top)^{-1} w^{\text{dem}}$ (a first-welfare-theorem for the wrench market);*
- (ii) *for the rigidly grasped load as a constrained Euler–Lagrange system, the Lagrange multiplier of the contact constraint under minimum-effort resolution in the metric R (Gauss’s principle) is exactly μ^* : the Nash price of the wrench is the generalized constraint force;*
- (iii) *at force saturation, the dual certificate of the membership program is a supporting normal of $\mathcal{W}(C)$ at w^{dem} : the price aligns with the facet the demand is about to violate—the minimizing direction of the margin in Proposition 2.*

Proof. (i) Stationarity of the Lagrangian $\frac{1}{2} \lambda^\top R \lambda - \mu^\top (G\lambda - w)$ gives $\lambda = R^{-1} G^\top \mu$ and $GR^{-1} G^\top \mu = w$, invertible by full row rank; the per-contact best response reproduces $\lambda_h = R_h^{-1} A_h^\top \mu$ and market clearing yields the same linear system; uniqueness by strict convexity. (ii) In quasistatics the contact rigidity constraint has Jacobian G ; the indeterminacy of λ on $\ker G$ resolved by minimum R -norm is problem (a), whose multiplier is μ^* . (iii) KKT of the membership LP at $\rho = 0$ produces η with $\eta^\top w = h_{\mathcal{W}}(\eta)$, a supporting hyperplane; LP duality identifies it with the limiting price. $\square$

Proposition 6 (Internal forces and grasp maintenance). *Decompose $\lambda = \lambda_{\text{mov}} + \lambda_{\text{int}}$ with $G\lambda_{\text{int}} = 0$. Then (i) internal (squeeze) forces have zero market price, $\mu^\top G\lambda_{\text{int}} = 0$, so grasp maintenance does not compete for motion authority; (ii) a rigidity-maintaining squeeze exists iff $(\lambda_{\text{mov}} + \ker G) \cap \prod_h \text{Cone}_h \neq \emptyset$, a feasibility LP whose Chebyshev slack is an online rigidity certificate.*

8 The control law: coupled strategy–motion dynamics

We now derive the control law and prove closed-loop convergence of the full coupling between strategic revision and physical motion, *without timescale separation*. The structural principle is that one potential generates both.

8.1 Body-frame load dynamics and the constant zonotope

Proposition 7 (Newton–Euler reduction). *Let $g = (R(\theta), p) \in \text{SE}(2)$ and body twist $\xi = (v_b, \omega) \in \mathbb{R}^3$. With contact forces expressed in the body frame,*

$$M_L \dot{\xi} + C_L(\xi)\xi = w_b + w_{\text{ext}}(g, \xi), \quad M_L = \text{diag}(m, m, J), \quad C_L(\xi) = \begin{bmatrix} m\omega S & 0 \\ 0 & 0 \end{bmatrix}, \quad (6)$$

with $S = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, and $w_b \in \mathcal{W}(C)$. Since slots and force sets are body-fixed, $\mathcal{W}(C)$ is constant along the motion, and $\xi^\top C_L(\xi)\xi = 0$ (workless gyroscopics).

Proof. Newton in the body frame, $m(\dot{v}_b + \omega S v_b) = f_b$, and planar Euler, $J\dot{\omega} = \tau_b$; the input set is the Minkowski sum of body-fixed images; $v_b^\top S v_b = 0$ by skew-symmetry. $\square$

Thus the object certifying recruitment (Proposition 2) is, without transformation, the input-constraint set of the load’s control problem. In particular $0 \in \text{int } \mathcal{W}(C) \iff \rho(C, 0) > 0$ iff the load is a *fully actuated* mechanical system, and for a desired trajectory $t \mapsto (g_d, \xi_d)$ the *demand tube* $w^{\text{dem}}(t) = M_L \dot{\xi}_d + C_L(\xi_d)\xi_d - w_{\text{ext}}(g_d, \xi_d)$ yields the *tube margin* $\rho_{\text{tube}}(C) = \min_t \rho(C, w^{\text{dem}}(t))$: the coalition can execute the plan with uniform authority reserve β iff $\rho_{\text{tube}} \geq \beta$, and the minimizing pair (t^*, η^*) diagnoses *where* the plan is fragile and *which* wrench direction (hence slot) to reinforce.

8.2 The consistent potential and the control law

Definition 2 (Strategic–mechanical potential; Nash–Lagrange equilibrium). *With state-dependent payoffs $p_{ik}(x, q) = V_k \sigma(D_k - S_k(x)) - \alpha V(q_i, p_k)$ ($V \geq 0$, C^1 distance potential), define*

$$\Psi(x, q) = \sum_k V_k \Psi_k(S_k(x)) - \alpha \sum_{i,k} c_i x_{ik} V(q_i, p_k), \quad (7)$$

so $\partial \Psi / \partial x_{ik} = c_i p_{ik}(x, q)$ and $\nabla_{q_i} \Psi = -\alpha c_i \sum_k x_{ik} \nabla_{q_i} V(q_i, p_k)$. A pair (x^*, q^*) is a Nash–Lagrange equilibrium (NLE) if $x^* \in \text{NE}(p(\cdot, q^*))$ and $\nabla_q \Psi(x^*, q^*) = 0$.

Control law. During the approach phase ($f_i = 0$), each robot applies

$$\tau_i = g_i(q_i) - K_i \dot{q}_i + \nabla_{q_i} \Psi(x, q) = g_i(q_i) - K_i \dot{q}_i - \alpha c_i \sum_k x_{ik} \nabla_{q_i} V(q_i, p_k), \quad K_i \succ 0, \quad (8)$$

a gravity-compensated, damped descent of the *preference-weighted attraction field*: motion follows the same potential that drives strategy revision, and the coalition literally emerges while the robots move. Upon contact, forces are allocated by the market law $\lambda = R^{-1} G^\top \mu$, $\mu = (GR^{-1} G^\top)^{-1} w_{\text{cmd}}$ of Theorem 6, with w_{cmd} from a geometric PD+feedforward tracking law on $\text{SE}(2)$ [18], projected onto $\mathcal{W}_{\text{safe}}$ (Section 8.5) when saturated.

Theorem 7 (Closed-loop convergence to Nash–Lagrange equilibria). *Under Assumptions 1–3, consider the coupled system: Smith revision (4) with payoffs $p_{ik}(x(t), q(t))$, and Euler–Lagrange motion under (8). Then the total energy*

$$E(x, q, \dot{q}) = \sum_i \frac{1}{2} \dot{q}_i^\top M_i(q_i) \dot{q}_i + [\bar{\Psi} - \Psi(x, q)], \quad \bar{\Psi} = \sup_{\mathcal{X} \times \mathcal{Q}} \Psi < \infty,$$

satisfies along trajectories

$$\dot{E} = - \sum_i \dot{q}_i^\top K_i \dot{q}_i - \sum_i c_i \sum_{k,\ell} x_{i\ell} ([p_{ik} - p_{i\ell}]_+)^2 \leq 0, \quad (9)$$

and every trajectory converges to the set of Nash–Lagrange equilibria (with $\dot{q} \rightarrow 0$). The NLE set is nonempty.

Proof. Well-posedness. $\mathcal{X} \times \mathcal{Q}$ compact, Ψ continuous, so $\bar{\Psi} < \infty$ and $E \geq 0$; the closed-loop field is locally Lipschitz; simplex invariance as in Theorem 3; q remains in the compact workspace.

Dissipation. Along (2) with $f_i = 0$ and using skew-symmetry of $\dot{M}_i - 2C_i$, $\frac{d}{dt}(\frac{1}{2}\dot{q}_i^\top M_i \dot{q}_i) = \dot{q}_i^\top (\tau_i - g_i) = \dot{q}_i^\top (-K_i \dot{q}_i + \nabla_{q_i} \Psi)$. Also $\frac{d}{dt}[-\Psi] = -\nabla_x \Psi^\top \dot{x} - \nabla_q \Psi^\top \dot{q}$. Summing, the cross terms $\pm \nabla_q \Psi^\top \dot{q}$ cancel *exactly*—this is the role of consistency between the payoff’s cost term and the control field in (8)—leaving $\dot{E} = -\sum_i \dot{q}_i^\top K_i \dot{q}_i - \sum_i c_i(p_i \cdot \dot{x}_i)$, and Lemma 2 (valid pointwise for the instantaneous payoffs, its proof being algebraic in $(p, \dot{x})$) gives (9).

Invariance. By LaSalle on the compact invariant domain, trajectories converge to the largest invariant set in $\{\dot{q} = 0\} \cap \{\text{Smith at rest}\}$. Smith at rest means $x \in \text{NE}(p(\cdot, q))$ (Lemma 2); $\dot{q} \equiv 0$ invariant implies $\ddot{q} = 0$, and (8) in (2) forces $\nabla_{q_i} \Psi = 0$ for all i : precisely Definition 2.

Existence. A global maximizer of the continuous Ψ over the compact $\mathcal{X} \times \mathcal{Q}$ satisfies both first-order conditions (interior q for a sufficiently large workspace; maximization of the weighted potential over x implies Nash). $\square$

Remark 4. (i) *There is no timescale separation and no smallness condition: one Lyapunov function certifies the interconnection.* (ii) *If the motion controller uses a different potential $\tilde{V}$ (e.g., a planner with obstacle shaping) with $\|\nabla \tilde{V} - \nabla V\| \leq \delta$, the cross terms no longer cancel and a standard singular-perturbation argument yields practical convergence to an $O(\delta + \epsilon)$ neighborhood under slow revision $\dot{x} = \epsilon(\cdot)$ [20].* (iii) *The theorem holds verbatim for any positive-correlation revision protocol and extends to the marginal-capacity payoffs of Theorem 2 with e_{ik} frozen at events.*

8.3 Margin-certified tracking

Theorem 8 (Tracking with market allocation). *Let w_{cmd} be a geometric PD+feedforward law on SE(2) for the fully actuated load (6) [18], allocated by Theorem 6. If $w_{\text{cmd}}(t) \in \mathcal{W}(C) \ominus \mathbb{B}_{\beta'}$ along the trajectory (guaranteed when the PD correction is bounded by the tube reserve β of the recruitment certificate), the pose error converges exponentially with a region of attraction quantified by ρ_{tube} ; under saturation, projection of w_{cmd} onto $\mathcal{W}(C)$ yields an input-to-state bound in $d(w_{\text{cmd}}, \mathcal{W}(C))$.*

Proof outline. Full actuation holds by Proposition 7 and $\rho > 0$; exponential tracking for fully actuated systems with the configuration-error framework is classical [18]; the allocation realizes any commanded $w_b \in \mathcal{W}(C)$ exactly (Theorem 6(i)); the saturated case treats the projection residual as a bounded disturbance in the ISS sense. $\square$

8.4 Distributed predictive transport is an exact dynamic potential game

Discretize (6) with shared load state $z_{t+1} = f(z_t, \sum_j G_j \lambda_{j,t})$; robot i plans $\lambda_i = (\lambda_{i,t})_{t < T} \in \mathcal{U}_i$, with cost

$$J_i(\lambda_i, \lambda_{-i}) = \sum_{t=0}^T \tilde{\ell}_c(z_t) + \sum_{t=0}^{T-1} \ell_i(\lambda_{i,t}), \quad (10)$$

where $\tilde{\ell}_c$ collects the shared tracking cost and all common state-dependent penalties, and ℓ_i the private effort.

Theorem 9 (Team-optimal distributed Nash). *The open-loop game (10) admits the exact potential $P(\lambda) = \sum_t \tilde{\ell}_c(z_t) + \sum_j \sum_t \ell_j(\lambda_{j,t})$: $J_i(\lambda'_i, \lambda_{-i}) - J_i(\lambda) = P(\lambda'_i, \lambda_{-i}) - P(\lambda)$ for every unilateral deviation. Hence every centralized minimizer is a Nash equilibrium; and in the convex case (LTV linearization, convex costs, polytopic $\mathcal{U}_i$) the variational GNE coincides with the centralized optimum: distributed KKT splitting (with monitorable primal–dual residuals) solves the team problem exactly. Moreover, the load costate induces, through the input map, a time-indexed price μ_t with stationarity $\nabla \ell_i(\lambda_{i,t}) = G_i^\top \mu_t$: the market law of Theorem 6 extended along the horizon.*

Proof. A deviation of i changes the shared trajectory term identically in J_i and P , and only its own effort otherwise; the identity follows. The convex-case statement is the coincidence of the variational KKT systems of the game and of $\min P$; the costate claim is Pontryagin stationarity in $\lambda_{i,t}$. $\square$

8.5 Safety: half-spaces in wrench space

Proposition 8 (HOCBF constraints are wrench half-spaces). *Let $h(g) \geq 0$ be a C^2 barrier on the transported body’s pose (signed distance of the swept body to an obstacle). On (6), h has relative degree two with respect to w_b , and the high-order barrier condition $\ddot{h} + k_1\dot{h} + k_0h \geq 0$ is affine in w_b : $b(g, \xi)^\top w_b \geq a(g, \xi)$ with closed-form a, b . Consequently the safe wrench set $\mathcal{W}_{\text{safe}}(g, \xi) = \mathcal{W}(C) \cap \bigcap \{w : b^\top w \geq a\}$ is a polytope, the support-function machinery of Proposition 2 applies unchanged, and a single geometric test $\rho_{\text{safe}}(t) = \rho(\mathcal{W}_{\text{safe}}, w_{\text{cmd}}(t))$ certifies capability and safety jointly. The safety-filter QP $\min_{w \in \mathcal{W}_{\text{safe}}} \|w - w_{\text{cmd}}\|^2$ produces multipliers $\nu \geq 0$ that enter the market as safety prices.*

Proof. $\dot{h} = \nabla h^\top J_g \xi$ is independent of w_b ; $\ddot{h}$ contains $\dot{\xi} = M_L^{-1}(w_b + w_{\text{ext}} - C_L \xi)$, affine in w_b ; collecting terms yields the half-space. Intersections of polytopes with half-spaces are polytopes. $\square$

The deliberative layer thus monitors five scalar certificates with physical semantics— Φ_d (assignment quality), ρ_{tube} (authority), ρ_{safe} (joint capability–safety), the grasp slack of Proposition 6, and the tracking Lyapunov function—and triggers re-recruitment (through the guard of Theorem 4, hence with finite events) when authority or safety prices spike. Formal verification of the architecture reduces to monitoring these functions: this is the operational meaning of *white-box*.

9 Endogenous communication topology

In a mobile fleet the communication graph $\mathcal{G}(q)$, its Laplacian $L(q)$ and Fiedler value $\lambda_2(L(q))$ are outcomes of the game, not assumptions. Information has an explicit exchange rate to physical units: composing the perturbation of the market solution with dynamic-consensus error bounds yields a decentralization overcost

$$\frac{J_{\text{dec}} - J^\star}{J^\star} \leq \kappa (GR^{-1}G^\top) \left(\frac{c_0 \dot{s}_{\text{max}}}{\|w_{\text{dem}}\|} \right)^2 \min \left\{ \lambda_2(L(q))^{-2}, \tau_{\text{max}}^2 \right\} + o(\cdot), \quad (11)$$

where τ_{max} is the maximal age of information and $\dot{s}_{\text{max}}$ the signal rate [22, 23]: relaying buys watt-hours in bounded, computable quantity.

Proposition 9 (Free riding). *Connectivity is a public good: with private relaying cost $\kappa > 0$ and common benefit B shared as $b = B/N < \kappa < B$ in individual payoffs, “no one relays” is the unique Nash equilibrium while the welfare optimum relays: undesigned endogenous topology is structurally under-provided. Including the common information value $c_\lambda \Lambda(q)$ (e.g., $\phi(\lambda_2(L(q)))$, concave) in every payoff—the same shared+private structure as Theorem 9—restores exact potential structure $\Psi_{\text{ext}} = \Psi + c_\lambda \Lambda$ and internalizes the externality; Theorem 7 extends where λ_2 is simple (differentiable, with distributed gradient $\partial \lambda_2 / \partial a_{ij} = (v_i - v_j)^2$ [24]), with Clarke-nonsmooth analysis at eigenvalue crossings.*

Theorem 10 (Comparative-advantage specialization and its selection). *Consider the role game: capacities $c_1 \leq \dots \leq c_N$; transporting pays a_{c_i} (marginal-capacity pricing, Theorem 2); relaying pays $\nu(r)$, strictly decreasing in the number r of active relays and independent of c_i , funded by (11). Then:*

- (i) the welfare $W(\mathcal{S}) = \sum_{i \notin \mathcal{S}} ac_i + \sum_{j \in \mathcal{S}} \nu(j)$ is an exact potential of the game, and its global maximizers are exactly the sorted-threshold profiles: relays are the r^* lightest robots, with r^* the largest r such that $\nu(r) \geq ac_r$;
- (ii) sorted-threshold profiles are Nash equilibria; unsorted Nash equilibria may also exist under unilateral deviations (e.g., $\nu(r) \geq ac_i > ac_j \geq \nu(r+1)$ with a heavy relay i and light transporter j);
- (iii) however, every profile stable under pairwise exchanges is sorted-threshold: swapping a heavier relay with a lighter transporter keeps r fixed and increases the potential by $a(c_i - c_j) > 0$; since pairwise-comparison revision protocols (the Smith class) execute exactly such exchanges in the population limit, the evolutionary dynamics select the comparative-advantage equilibrium among the Nash set. Light robots become mobile hubs endogenously—not by programming, but because their opportunity cost in the capacity market is lowest.

Proof. Potential: a unilateral role switch of robot i changes its payoff by $\nu(r') - ac_i$, exactly ΔW . (i) For fixed r , maximizing W minimizes $\sum_{\mathcal{S}} c_i$, achieved by the r lightest (rearrangement); over r , $W(r) = \sum_i ac_i + \sum_{j \leq r} [\nu(j) - ac_j]$ has decreasing increments, hence is unimodal with the stated threshold. (ii) At a sorted threshold, no relay gains by transporting ($\nu(r^*) \geq ac_{r^*} \geq ac_i$ for relays) and no transporter by relaying ($ac_j \geq ac_{r^*+1} \geq \nu(r^*+1)$); the unsorted example in the statement verifies both unilateral conditions, proving non-uniqueness. (iii) The swap changes only the identities, so $\Delta W = a(c_i - c_j) > 0$; a profile admitting an improving swap is not pairwise-stable; conversely sorted-threshold admits none. $\square$

Remark 5. Part (iii) is an equilibrium selection result: the same pairwise-comparison structure that gives Smith dynamics positive correlation makes them discard the unsorted Nash equilibria that best-response reasoning tolerates. The fully endogenous fixed point (graph shapes information, information shapes payoffs, payoffs shape motion, motion shapes the graph) is posed as an open problem in Section 11.

10 Numerical validation of falsifiable predictions

The theory was validated in three audited Monte Carlo campaigns (reproducible pipelines; frozen, disjoint seeds for training/tuning/evaluation; paired hypothesis tests; theory-audit gates derived from the proofs; 53,580 checks in total with *zero* audit failures). We report the three experiments that test *predictions made by the theorems above*, not merely rankings.

(a) Marginal-capacity pricing (Theorem 2). With pair-dependent effective capacities, the theory predicts that the plain-payoff Smith rule ascends no potential of the game and should disperse capacity across incomplete loads, and that the marginal payoff (3) should repair this. In the dedicated ablation (9,360 checks, 1,560 paired runs, identical worlds), the marginal correction reduced the Smith score gap from 0.372 to 0.157, raised coalition success from 0.051 to 0.135, reduced the capacity trapped in incomplete loads from 0.946 to 0.848, and improved served-capacity alignment from 0.049 to 0.137, all with $p \ll 10^{-3}$: the theory detected a misalignment, prescribed the payoff, and the experiment confirmed the direction of every predicted effect.

(b) Insufficiency of scalar criteria and complementarity (Propositions 3–4, Theorem 5). In the wrench campaign (100 seeds, six geometric scenario families including pure-torque bars and one-sided pushing), scalar-capacity assignment produced physically infeasible coalitions at effect size 0.50 ($p = 1.6 \times 10^{-92}$), instantiating Proposition 3; the *marginal-margin* ($\Delta\rho$) signal underperformed the *deficit-margin* signal (gap 0.140 vs 0.112; the corresponding superiority hypothesis was *not* rejected), exactly the complementarity myopia predicted by Proposition 4; and an auction engine equipped with the *continuous* wrench signal attained near-oracle gap (0.016), confirming Remark 3: the impossibility of Theorem 5 belongs to the indicator, and the margin dissolves it—the signal is portable across engines.

(c) **Quality–resource separation at the recruitment layer.** Across 27,300 audited evaluations, the exact centralized oracle, a learned CTDE policy (4,674 parameters, gap 0.004, runtime 11.2 ms) and compact distributed rules (zero parameters, runtimes 0.1–0.5 ms, gaps 0.28–0.34) trace the quality–resource Pareto frontier that motivates the architecture: interpretable distributed rules with certified structure occupy the low-resource regime, and the theory of Sections 3–5 explains *which* structural ingredients (capacity weighting, marginal pricing, potential-guarded clearing) close part of the gap at negligible cost.

11 Limitations and open problems

The results are planar and assume omnidirectional contact forces, snapshot-frozen effective capacities, and known aggregates (Assumption 3); simulation validates predictions, not hardware. Four open problems define the research program:

Open Problem 1 (Nonholonomy). *Differential-drive robots have heading-dependent force cones $U_h(\theta_i)$, so $\mathcal{W}(C, \theta)$ is no longer constant. Characterize the largest fixed zonotope contained in $\bigcap_{\theta} \mathcal{W}(C, \theta)$ and its margin, or treat headings as strategic variables.*

Open Problem 2 (Contact uncertainty). *Friction and slip erode $\mathcal{W}$; robust tube margins $(\mathcal{W} \ominus \mathbb{B}_{\delta_w})$ preserve the polytopic machinery, and ISS tracking bounds must be closed.*

Open Problem 3 (Recursive feasibility of the hybrid predictive game). *Prove recursive feasibility and stability of the receding-horizon GNE with terminal sets defined by residual wrench authority and with discrete re-recruitment events governed by the potential guard.*

Open Problem 4 (Fully endogenous topology). *Existence and convergence for the fixed point topology $\leftrightarrow$ strategy, including joint role/patrol/coalition games and the characterization of the optimal hub count $r^*(a, \nu, \dot{s}_{\max})$.*

12 Conclusion

A single geometric object—the constant body-frame reachable-wrench zonotope—and a single economic object—the wrench price—thread recruitment, grasping, motion, prediction and safety into one theory with proofs at every junction: an iff characterization of potential structure under heterogeneity; finite-event hybrid convergence with explicit budgets; impossibility results that locate exactly where head-count, scalar capacity and DMG auctions fail, together with the continuous margin that dissolves the impossibility; a triple identity unifying market, mechanics and geometry; a control law certified by one energy function for the strategy–motion interconnection; team-optimal distributed predictive transport; safety as wrench half-spaces; and an equilibrium-selection account of why the market staffs mobile communication hubs with its lightest robots. Every guarantee is a monitorable scalar with physical semantics—the operational content of formally verifiable, white-box distributed autonomy.

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